# Trusted Data Foundation: Architecture Proposal

Babak Bashiri · Data platform proposal for a global industrial manufacturer · October 2026

**Runnable demo:** github.com/babay-bash/manufacturing-data-demo **Offline macOS app (Apple Silicon):** download, unpack, run `./start.command`

**Recommendation**

Build a trusted data foundation before expanding AI. Start with one business process, its key sources and a few agreed metrics, shared by dashboards, SQL and AI.

The problem is not tools. It is unclear ownership, conflicting KPIs, uneven quality, and AI answers that can't be traced.

## Ingestion

- CDC where reliable; APIs or exports otherwise.
- Keep source IDs, event and ingestion time, schema version.
- Kafka for telemetry where volume justifies it.
- Immutable archive for replay; dedup on event IDs.

## Storage and processing

- Snowflake + dbt by default; reuse Databricks if it is already in place.
- Raw stays unchanged; dbt standardizes.
- Facts with explicit grain, shared dimensions.
- Crosswalk to canonical IDs, rules agreed with owners.

## Metrics and access

- Each metric: owner, definition, grain, time rules.
- One name, one meaning; CI rejects conflicts.

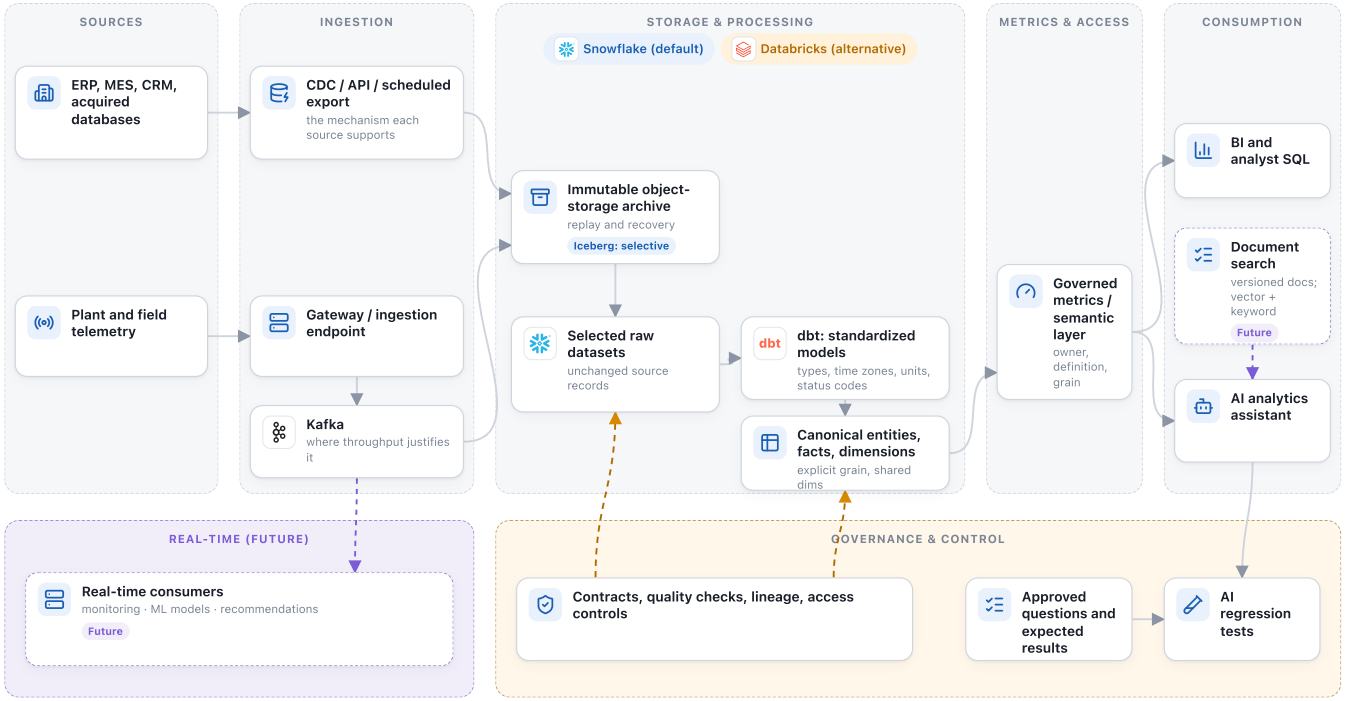
### Security model

Access requirements from discovery are translated into:

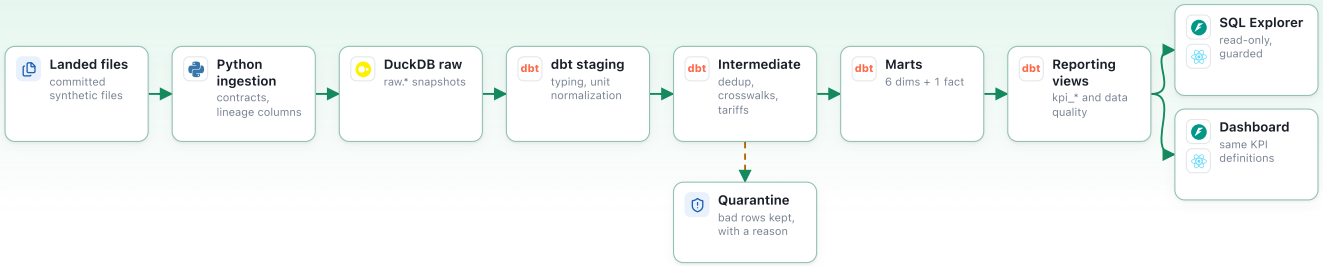
- Access and functional roles, with RBAC in the database.
- Row-level security (e.g. facility isolation); column masking for sensitive fields.
- SQL, BI and AI carry the user's identity end to end, so every query goes through database RBAC.

## Proposed architecture

Proposal · Orange dashed = governance · Purple dashed = future




## What this demo actually runs




Served by **FastAPI**, shown by a **React** app. Not deployed locally: **Kafka, object storage, Snowflake, CDC, AI**.

### Assumptions

- No reliable enterprise-wide customer, supplier or asset model today.
- Some sources support CDC; others only APIs or files.
- Not every telemetry signal needs real-time analytics.
- Small team; no need to migrate every pipeline at once.

### Questions for the client

1. Which key KPIs does the business depend on, and which cause the most pain today? Who owns each one?
2. What are the functional requirements for security and access: who should see which records, which facilities must be isolated, and which fields are sensitive?
3. For each priority source: which integration options exist, how often does data change, what are average and peak volumes, and what fault tolerance and failure modes must we handle? Which use cases truly need real-time data?
4. Which platforms and controls already exist that we should reuse, and which systems are authoritative for customer, supplier, asset and facility identity?
5. Which AI use cases are most urgent, and what counts as a correct, traceable answer?

## Where I would go deeper

### 1. Governed KPI example

- Pick one disputed KPI with a business owner.
- Agree numerator, denominator, grain, exclusions.
- Reconcile with the business, then build tested dbt models and one certified definition.
- Dashboard SQL and AI return the same number.

### 2. Quality and AI correctness

- Check platform health, pipeline delivery and data correctness.
- Deterministic checks before anomaly detection.
- Test fixtures with known answers; approved AI questions with expected results.
- Log questions, generated SQL and freshness for review.

## Choices and tradeoffs

| | | |
|---|---|---|
| **Focused MVP** | over enterprise rebuild | Prove one trusted path first. |
| **Reuse Snowflake / Databricks** | over platform replacement | Migration needs a measured benefit. |
| **Raw archive** | over all telemetry in Snowflake | Replay without processing unused signals. |
| **Selective Iceberg** | over a blanket rule | Worth it only when several engines share large data. |
| **Shared definitions** | over competing gold tables | Variants stay explicit and named. |
| **Measured AI evals** | over prompt-only fixes | Prompts can’t prove correctness. |

## First 90 days

DAYS 1–30

### Agree and baseline

- Inventory sources and competing KPIs
- Pick the key KPIs and an AI question set
- Name owners; confirm access, volume and latency risks

DAYS 31–60

### Build one end-to-end path

- Land sources; approve KPI SQL
- Staging, facts/dims, dbt tests, semantic definition
- CI, delivery checks, lineage, alerting

DAYS 61–90

### Certify and demonstrate

- Reconcile the KPIs with their owners
- Same result in SQL, dashboard and AI
- Report freshness, failures and limits

### Team of four

- **Lead (me):** architecture, stakeholders, scope, AI correctness
- **Ingestion:** sources and reconciliation
- **Analytics engineering:** dbt models and semantic definitions
- **Platform:** CI, access, monitoring, AI integration

### Definition of done

- A key set of business-approved KPIs, each with an owner and lineage
- Tested models and agreed freshness
- Same answer in SQL, dashboard and AI
- Access controls, repeatable deployment, visible failures

## Future evolution

### Add KPIs and facts iteratively

Iterative

1. Start from a KPI with an owner.
2. Ingest only the new sources it needs.
3. Reuse conformed dimensions; extend them first.
4. Add a dimension only for a new entity needed for slicing or a feature store.
5. Certify the KPI and add regression cases.

### Real-time use cases

Future

Kafka feeds these directly:

- Operational monitoring and alerting
- ML models (online features)
- Recommendation systems
- Only where seconds or minutes matter; batch stays the default.

### AI on documents (RAG)

Future

1. First: versioning. Owners mark docs current or outdated.
2. Latest valid version by default.
3. Then hybrid search: vector similarity + keyword.
4. Permission-filtered, with citations and dates.
5. Test that the right source ranks first.

### Not building yet

- Company-wide canonical model
- Wholesale dashboard migration
- Real-time for every source
- Broad document RAG
- Custom AI orchestration layer
- Iceberg everywhere

### AI use disclosure

- Write-up: AI helped organize, edit and challenge trade-offs. Priorities and final choices are mine.
- Demo, generated with AI (Claude Code) from my specification and reviewed by me: the application code, the synthetic data and source samples (MES, QMS, ERP, telemetry), and the manufacturing KPI definitions.

*A proposal based on stated assumptions; products and SLAs would be confirmed in discovery.*